

Airbus Product Standard
Standard L section, extruded
Aluminium alloy 7175
with peripheral coarse grain control
 $1,2\text{ mm} \leq a \leq 18,0\text{ mm}$

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1 Scope

This standard specifies the dimensions, tolerances and mass of extruded L sections in aluminium alloy 7175 for aerospace applications.

2 Normative References

This Airbus Standard incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any these publications apply to this Airbus Standard only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS03-05-000	Airbus Material Specification – Aluminium and aluminium alloy wrought products bar and section – Technical Specification
AIMS03-05-009	Airbus Material Specification – Aluminium alloy (7175). Solution treated, controlled stretched and artificially aged (T73511). Extruded section $1,2 \text{ mm} \leq a \leq 38,0 \text{ mm}$ with peripheral coarse grain control – Material Specification.
AIMS03-05-012	Airbus Material Specification – Aluminium alloy (7175). Solution treated and artificially aged (T79511). Extruded section $1,2 \text{ mm} \leq a \leq 38,0 \text{ mm}$ with peripheral coarse grain control – Material Specification.
AIMS03-05-019	Airbus Material Specification – Aluminium alloy (7175) solution treated and artificially aged (T73) extruded section $1,2 \leq a \leq 38,0 \text{ mm}$ with peripheral coarse grain control – Material Specification
DIN1748 Part 4	Aluminum and wrought aluminum alloy extruded profiles - Permissible Deviations
EN2066	Extruded section in aluminum and aluminum alloys, general tolerances

3 Form

See Figure 1.

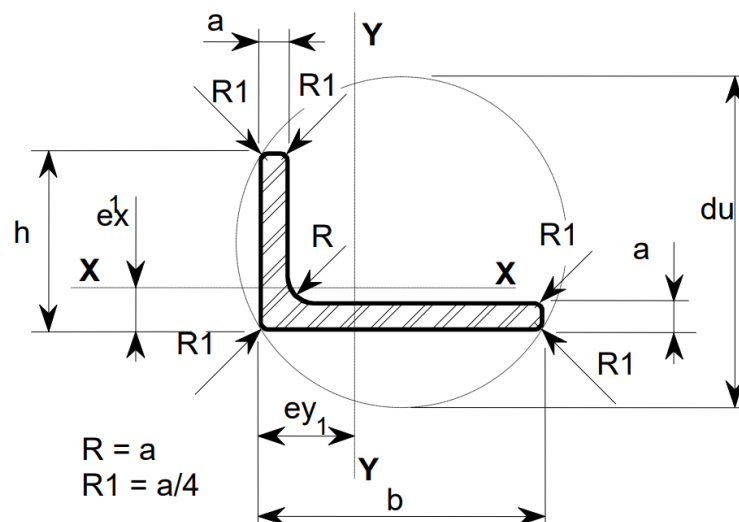


Figure 1: Configuration

4 Recommended Dimensions, Mass and Static Values

4.1 Dimensions and Mass

See Table 1.

Table 1: Dimensions and Mass

Dimensions in mm

Size Code	h	b	a	Ø du	$\frac{\text{Ø du}}{a_{\min}}$	Cross-section mm ²	Length	Linear mass ^{a)} kg/m
0036	10	10	2	14,14	7,07	36,60	Stock length not less than 2 000 and exact length by agreement	0,102
0046		15		18,03	9,01	46,60		0,130
0056		20		22,36	11,18	56,60		0,158
0027	12	12	1,2	16,97	14,14	27,58		0,077
0060		20	2	23,32	11,66	60,60		0,170
0043	15	15	1,5	21,21	14,14	43,09		0,121
0057			2		10,61	56,60		0,158
0076		25		29,15	14,58	76,60		0,214
0037	16	16	1,2	22,63	18,86	37,18		0,104
0049			1,6		14,14	49,02		0,137
0061			2		11,31	60,60		0,170
0041		20	1,2	25,61	21,34	41,98		0,118
0055			1,6		16,01	55,42		0,155
0068			2		12,81	68,60		0,192
0063		25	1,6	29,68	18,55	63,42		0,178
0079			2		14,84	78,60		0,220
0098			2,5		11,87	97,19		0,272
0059	18,8	20	1,6	27,32	17,08	60,06		0,168
0047	20	20	1,2	28,28	23,57	46,78		0,131
0058			1,5		18,86	58,09		0,163
0062			1,6		17,68	61,82		0,173
0077			2		14,14	76,60		0,214
0094			2,5		11,31	94,69		0,265
0070		25	1,6		20,01	70,00		0,196
0086			2		16,01	86,60		0,242
0107			2,5		12,81	107,19		0,300
(continued)								

(continued)

Table 1: Dimensions and Mass (continued)

Dimensions in mm

Size Code	h	b	a	Ø du	Ø du a _{min}	Cross-section mm ²	Length	Linear mass ^{a)} kg/m
0074	20	28	1,6	28,28	21,51	74,70	Stock length not less than 2 000 and exact length by agreement	0,209
0096		30	2		18,03	96,60		0,270
0142			3		12,02	142,35		0,399
0081		32	1,6	37,74	23,58	81,20		0,227
0100			2		18,87	100,60		0,282
0124			2,5		15,09	124,69		0,349
0148			3		12,58	148,35		0,415
0158			3,2		11,79	157,70		0,442
0116		40	2	44,72	22,36	116,60		0,326
0144			2,5		17,89	144,69		0,405
0183A			3,2		13,98	183,30		0,513
0078	25	25	1,6	35,36	22,10	78,00		0,218
0097			2		17,68	96,60		0,270
0105			2,2		16,07	105,88		0,296
0119			2,5		14,14	119,69		0,335
0151			3,2		11,05	151,30		0,424
0089		32	1,6	40,61	25,38	89,20		0,250
0102			1,8		22,56	102,00		0,285
0110			2		20,30	110,60		0,310
0137			2,5		16,24	137,19		0,384
0173			3,2		12,69	173,70		0,486
0157A		40	2,5	47,17	18,87	157,19		0,440
0187			3		15,72	187,35		0,525
0199			3,2		14,74	199,30		0,558
0246		40	4	47,17	11,79	246,40		0,690
0183		50	2,5	55,90	22,36	183,00		0,512
0231			3,2		17,47	231,30		0,648
0286			4		13,98	286,40		0,802
0145	30	30	2,5	42,43	16,97	144,69		0,405

(continued)

Table 1: Dimensions and Mass (continued)

Dimensions in mm

Size Code	h	b	a	Ø du	Ø du a _{min}	Cross-section mm ²	Length	Linear mass ^{a)} kg/m
0172	30	30	3	42,43	14,14	172,35	Stock length not less than 2 000 and exact length by agreement	0,483
0236		32,5	4	44,23	11,06	236,40		0,662
0232		50	3	58,31	19,44	232,35		0,651
0306			4		14,58	306,40		0,858
0125	32	32	2	45,25	22,63	125,00		0,350
0154			2,5		18,10	154,69		0,433
0184			3		15,08	184,35		0,516
0196			3,2		14,14	196,10		0,549
0175		40	2,5	51,22	20,49	175,00		0,490
0221			3,2		16,01	221,70		0,621
0274			4		12,81	274,40		0,768
0253		50	3,2	59,36	18,55	253,70		0,710
0314			4		14,84	314,40		0,880
0388			5		11,87	388,75		1,089
0295		63	3,2	70,66	22,08	295,30		0,827
0366			4		17,67	366,40		1,026
0453			5		14,13	453,75		1,271
0153	37	41	2	54,81	27,41	152,59		0,427
0157	40	40	2	56,57	28,28	157,00		0,440
0195			2,5		22,63	195,00		0,546
0233			3		18,86	232,35		0,651
0247			3,2		17,68	247,30		0,692
0307			4		14,14	306,40		0,858
0279		50	3,2	64,03	20,01	279,30		0,782
0346			4		16,01	346,40		0,970
0429			5		12,81	429,00		1,201
0386		60	4	72,11	18,03	386,40		1,082
0478			5		14,42	478,75		1,341

(continued)

Table 1: Dimensions and Mass (continued)

Dimensions in mm

Size Code	h	b	a	Ø du	$\frac{\text{Ø du}}{a_{\min}}$	Cross-section mm ²	Length	Linear mass ^{a)} kg/m
0398	40	63	4	74,63	18,66	398,40	Stock length not less than 2 000 and exact length by agreement	1,116
0493			5		14,93	493,75		1,383
0587			6		12,44	587,40		1,645
0466		80	4	89,44	22,36	466,40		1,306
0578			5		17,89	578,75		1,621
0722			6,3		14,20	722,26		2,022
0245	50	50	2,5	70,71	28,28	245,00		0,686
0312			3,2		22,10	312,00		0,874
0387			4		17,68	386,40		1,082
0479			5		14,14	478,75		1,341
0438		63	4	80,43	20,11	438,40		1,228
0543			5		16,09	543,75		1,523
0678			6,3		12,77	678,16		1,899
0628		80	5	94,34	18,87	628,75		1,761
0749			6		15,72	749,40		2,098
0785			6,3		14,97	785,26		2,199
0985			8		11,79	985,60		2,760
0728		100	5	111,80	22,36	728,75		2,041
0869			6		18,63	869,40		2,434
0911			6,3		17,75	911,26		2,552
1145			8		13,98	1145,60		3,208
0579	60	60	5	84,85	16,97	578,75		1,621

(continued)

Table 1: Dimensions and Mass (concluded)

Dimensions in mm

Size Code	h	b	a	Ø du	Ø du a _{min}	Cross-section mm ²	Length	Linear mass ^{a)} kg/m
0689	60	60	6	84,85	14,14	689,40	Stock length not less than 2 000 and exact length by agreement	1,930
0396		75	3	96,12	32,04	396,00		1,108
0491	63	63	4	89,10	22,27	491,00		1,370
0608			5		17,82	608,75		1,705
0725			6		14,85	725,40		2,031
0760			6,3		14,14	760,06		2,128
0693		80	5	101,83	20,37	693,75		1,943
0867			6,3		16,16	867,16		2,428
1089			8		12,73	1089,60		3,051
0993		100	6,3	118,19	18,76	993,16		2,781
1249			8		14,77	1249,60		3,499
1545			10		11,82	1545,00		4,326
0780	80	80	5	113,14	22,63	780,00		2,180
0929			6		18,86	929,40		2,602
0974			6,3		17,96	974,26		2,728
1225			8		14,14	1225,60		3,432
2449		100	13,5	128,06		2449,00		6,857
1100			6,3		20,33	1100,26		3,081
1385			8		16,01	1385,60		3,880
1715			10		12,81	1715,00		4,802
1965		125	6,3	148,41	14,84	1965,00		5,502
2337			12		12,37	2337,60		6,545
1546	100	100	8	141,42	17,68	1545,60		4,328
1915			10		14,14	1915,00		5,362
2997		160	12	188,68	15,72	2997,60		8,393
3942			16		11,79	3942,40		11,039
6743	160	230	18	276,50	15,36	6743,00		1,880

^{a)} For information, calculated with a density of 2,8 kg/dm³.

4.2 Static Values

See Table 2.

Table 2: Static Values

Dimensions in mm

Size code	ex ₁	ey ₁	lxx mm ⁴	lyy mm ⁴
0027	3,44	3,44	373	373
0036	3,22	3,22	314	314
0037	4,44	4,44	917	917
0041	4,00	6,00	980	1704
0043	4,30	4,30	911	911
0046	2,74	5,24	356	1009
0047	5,16	5,16	1834	1834
0049	4,59	4,59	1180	1180
0055	4,15	6,15	1262	2205
0056	2,43	7,43	384	2264
0057	4,48	4,48	1160	1160
0058	5,56	5,56	2239	2239
0059	5,11	5,70	2333	1998
0060	3,00	7,00	660	2420
0061	4,73	4,73	1424	1424
0062	5,59	5,59	2371	2371
0063	3,72	8,22	1342	4089
0068	4,29	6,29	1525	2677
0070	7,50	5,00	4428	2540
0074	8,70	4,70	6025	2604
0076	3,57	8,57	1345	4875
0077	5,74	5,74	2880	2880
0078	6,84	6,84	4738	4738
0079	3,87	8,37	1623	4987
0081	10,40	4,40	8733	2714
0086	5,19	7,69	3082	5384
0089	9,50	6,10	9405	5108
0094	5,92	5,92	3476	3476
0096	4,75	9,75	3242	8922
0097	6,99	6,99	5788	5788
0098	4,06	8,56	1948	6046
0100	4,60	10,60	3297	10657
0102	6,12	9,59	5682	10488

(continued)

Table 2: Static Values (Continued)

Dimensions in mm

Size code	ex ₁	ey ₁	lxx mm ⁴	lyy mm ⁴
0105	7,06	7,06	6294	6294
0107	5,37	7,87	3723	6535
0110	6,23	9,73	6231	11498
0116	4,10	14,10	3481	19641
0119	7,17	7,17	7031	7031
0124	4,79	10,79	3986	13003
0125	8,70	8,70	12483	12483
0137	6,41	9,91	7575	14042
0142	5,12	10,12	4551	12721
0144	4,29	14,29	4212	24056
0145	8,42	8,42	12442	12442
0148	4,97	10,97	4631	15223
0151	7,42	7,42	8654	8654
0153	9,47	11,46	20 047	25 828
0154	8,92	8,92	15214	15214
0157	10,70	10,70	24820	24820
0157A	5,75	13,25	8050	26019
0158	5,04	11,04	4877	16094
0172	8,61	8,61	14580	14580
0173	6,67	10,17	9333	17402
0175	12,00	8,00	28402	16300
0183	17,50	5,10	48235	8506
0183A	4,56	14,56	5160	29930
0184	9,11	9,11	17853	17853
0187	5,94	13,44	9405	30623
0195	10,90	10,90	30476	30476
0196	9,18	9,18	18875	18875
0199	6,01	13,51	9927	32413
0221	8,30	12,30	20196	35286
0231	5,40	17,90	10490	59993
0232	6,76	16,76	16868	60358
0233	11,11	11,11	35818	35818
				(continued)

Table 2: Static Values (Continued)

Dimensions in mm

Size code	ex ₁	ey ₁	lxx mm ⁴	lyy mm ⁴
0236	8,66	9,91	19032	23244
0245	13,40	13,40	60597	60597
0246	6,30	13,80	11907	39292
0247	11,18	11,18	37932	37932
0253	7,45	16,45	21477	65432
0274	8,59	12,59	24397	42829
0279	10,08	15,08	40560	70577
0286	5,70	18,20	12597	73072
0295	6,62	22,12	22733	123301
0306	7,13	17,13	21520	78000
0307	11,47	11,47	46081	46081
0312	13,60	13,60	76027	76027
0314	7,74	16,74	25963	79795
0346	10,37	15,37	49307	86147
0366	6,92	22,42	27503	150976
0386	9,50	19,50	51872	142752
0387	13,98	13,98	92608	92608
0388	8,11	17,11	31164	96744
0396	14,39	21,85	67742	298635
0398	9,27	20,77	52543	163288
0429	15,60	10,70	104095	59039
0438	12,55	19,05	99249	176144
0453	7,30	22,80	33050	183942
0466	8,21	28,21	55703	314263
0478	9,87	19,87	62700	173950
0479	14,34	14,34	112503	112503
0491	17,10	17,10	190241	190241
0493	9,64	21,14	63621	199163
0543	12,92	19,42	120656	215036
0578	8,59	28,59	67394	384894

(continued)

Table 2: Static Values (concluded)

Dimensions in mm

Size code	ex ₁	ey ₁	lxx mm ⁴	lyy mm ⁴
0579	16,85	16,85	199078	199078
0587	10,01	21,51	73784	233254
0608	17,60	17,60	231774	231774
0628	11,50	26,50	128802	416302
0678	13,39	19,89	146722	262834
0689	17,21	17,21	233286	233286
0693	15,74	24,24	248943	450351
0722	9,08	29,08	81513	472352
0725	17,96	17,96	271889	271889
0728	10,26	35,26	135993	767243
0749	11,87	26,87	150484	489964
0760	18,07	18,07	283560	283560
0780	21,70	21,70	487716	487716
0785	11,98	26,98	159761	511480
0867	16,22	24,72	304794	553852
0869	10,64	35,64	158975	905775
0911	10,75	35,75	165638	946428
0929	22,22	22,22	573091	573091
0974	22,33	22,33	598506	598506
0985	12,61	27,61	190510	628670
0993	14,55	33,05	323974	1026531
1089	16,83	25,33	373050	681600
1100	20,12	30,12	639939	1113251
1145	11,39	36,39	201546	1169146
1225	22,95	22,95	737298	737298
1249	15,18	33,68	396854	1269684
1385	20,74	30,74	788913	1378353
1545	15,91	34,41	476118	1539221
1546	27,96	27,96	1481725	1481725
1715	21,47	31,47	952990	1672990
1915	28,68	28,68	1800044	1800044
1965	19,36	41,86	1014199	3117011
2337	20,09	42,59	1178028	3650013
2449	22,51	32,31	635509	2688173
2997	23,74	53,74	2407738	7839418
3942	25,21	55,21	3048164	10058724
6743	39,21	73,85	8090618	42179494

5 Tolerances

5.1 Dimensional Tolerances

See Table 3.

5.2 Geometrical Tolerances

See Table 3.

Table 3: Dimensional and Geometrical Tolerances

Standard	Tolerance Class	Code	
EN2066	Standard	E	Use only for order designation
		N	
DIN1748 Part 4	Precision	F	If this class is required, the code letter F has to be used for standard designation and order designation

6 Material

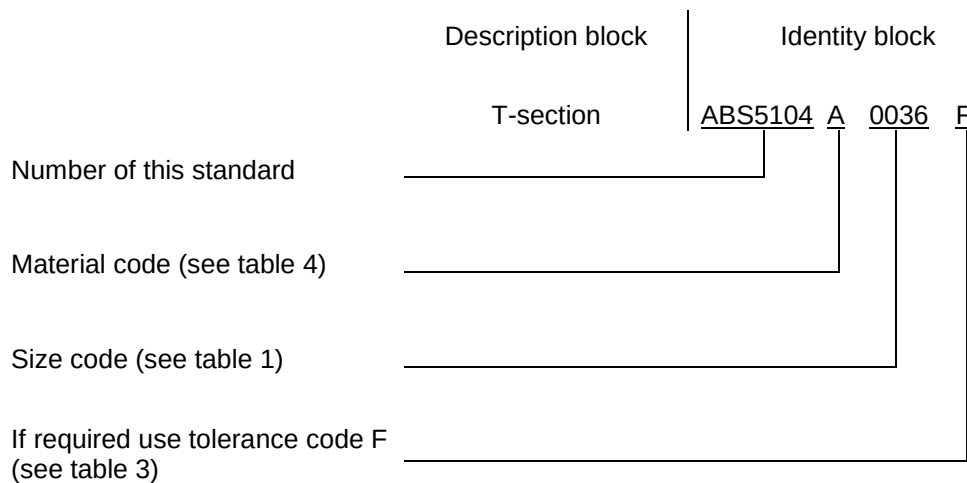
See Table 4.

Table 4: Material and condition of treatment

Thickness range	Material					
	Delivered			Use		
	AIMS spec. No.	Condition of treatment	Code	AIMS spec. No.	Condition of treatment	Code
1,2 mm ≤ a ≤ 38,0 mm	03-05-009	T73511	A	03-05-009	T73511	A
	03-05-019	O	E	03-05-019	T73	B
		F	G			
	03-05-012	T79511	C	03-05-012	T79511	C

7 Designation

This type of standard shall be designated according to the philosophy of the following example:



8 Technical Specification

AIMS03-05-000.

Record of Revisions

Issue	Clause modified	Description of Modification
1 05.1996	-	New Standard
2 09.2000	-	Size code 0102 added. For size code 0232 "a" = 4 changed to "a" = 3.
3 09.2002	-	Size code 6743 added. "1,2 mm ≤ a ≤ 100,0" changed to "1,2 mm ≤ a ≤ 38,0 mm". T79511 and T792 added.
4 09.2005	-	Template updated. Size codes 0396 and 2449 added. AIMS03-05-012, AIMS03-05-019 and AIMS03-05-020 added in paragraph "Normative references". Condition of treatment "O" and "F" added in table 4.
5 01.2013	-	Size code 0059 added.
6 11.2022	All Table 1+2 Table 2	Template updated. Size code 0153 added. Columns ex ₁ and ey ₁ transposed.